

MATH5702M Statistical Computing

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2026-09-28

Table of contents

About MATH5702	3
Organisation of MATH5702	3
Lectures	3
Problem sheets and problem classes	4
Coursework	5
Office hours	5
Exam	5
Content of MATH5702	6
Necessary background	6
Syllabus	7
Book	7
 I Monte Carlo estimation	 9
1 Introduction to Monte Carlo	10
2 Uses of Monte Carlo	11
3 Monte Carlo error I: theory	12
4 Monte Carlo error II: practice	13
5 Control variate	14
6 Antithetic variables I	15
Problem Sheet 1	16

About MATH5702

Organisation of MATH5702

This module is **MATH5702M Statistical Computing**.

This module lasts for 11 weeks from 29 September to 10 December 2026. The exam will take place sometime between 11 and 22 January 2027.

The module leader, the lecturer, and the main author of these notes is [Dr Matthew Aldridge](#). (You can call me “Matt”, “Matthew”, or “Dr Aldridge”, pronounced “*old*-ridge”.) My email address is m.aldridge@leeds.ac.uk, although I much prefer questions in person at office hours (see below) rather than by email.

The HTML webpage is the best way to view the course material. There is also a **PDF version**, although I have been much less careful about the presentation of this material, and it does not include the problem sheets.

Lectures

The main way you will learn new material for this module is by attending lectures. There are three lectures per week:

- Tuesdays at 1500 in 28 University Road, G.23
- Thursdays at 1100 in 28 University Road, G.23
- Thursday at 1400 in Roger Stevens LT17.

28 University Road can be tricky to find if you’ve not been there before: [here is a Google Maps link](#); [here is a video](#); [here is a map I drew in Powerpoint](#).

I recommend taking your own notes during the lecture. I will put brief summary notes from the lectures on this website, but they will not reflect all the details I say out loud and write on the whiteboard. Lectures will go through material quite quickly and the material may be quite difficult, so it’s likely you’ll want to spend time reading through your notes after the lecture. Lectures should be recorded on the lecture capture system; I find it very difficult to read the whiteboard in these videos, but if you unavoidably miss a lecture, for example due to illness, you may find they are better than nothing.

In Weeks 3, 5, 7, 9 and 11, the Thursday @ 11 lecture will operate as a problems class – see more on this below.

Attendance at lectures is compulsory. You should record your attendance using the UniLeeds app and the QR code on the wall in the 15 minutes before the lecture or the 15 minutes after the lecture (but not during the lecture).

Device policy in lectures:

- Please do not use phones in the lecture, unless taking part in a poll/survey or some other activity.
- Please only use a tablet/iPad in lectures if using it to make notes on the lecture.
- **New rule for 2026:** *Please do not use laptops during the lecture*
 - If you need a laptop for accessibility reasons, this is OK, but please let me know (by email or before/after a lecture).
 - If you are the world's fastest LaTeX-typist and you make notes on the lecture on your laptop, this is OK, but please let me know (by email or before/after a lecture).
 - I will hold a poll on this after two weeks – if you all hate the policy, I'll change it back.

(In the *problems classes* use of devices is fine – even encouraged.)

Problem sheets and problem classes

Mathematics and statistics are “doing” subjects! To help you learn material for the module and to help you prepare for the exam, I will provide 5 unassessed problem sheets. These are for you to work through in your own time to help you learn; they are *not* formally assessed. You are welcome to discuss work on the problem sheets with colleagues and friends, although my recommendation would be to write-up your “last, best” attempt neatly by yourself.

There will be an optional opportunity to submit one or two questions from the problem sheet to me in advance of the problems class for some brief informal feedback on your work. See the problem sheets for details.

You should work through each problem sheet in preparation for the problems class in the Thursday @ 11 lecture of Week 3, 5, 7, 9 and 11. In the problems class, you should be ready to explain your answers to questions you managed to solve, discuss your progress on questions you partially solved, and ask for help on questions you got stuck on.

You can also ask for extra help or feedback at office hours (see below).

Coursework

There will be one piece of assessed coursework, which will make up 20% of your module mark. [You can read more about the coursework here.](#)

The coursework will be in the form of a worksheet. The worksheet will have some questions, mostly computational but also mathematical, and you will have to write a report containing your answers and computations.

The assessed coursework will be introduced in the **computer practical** sessions later in the semester.

The deadline for the coursework will be the last-but-two day of the Autumn term, *Wednesday 9 December* at 1400. Feedback and marks will be returned on Monday 11 January 2027, the first day of the Spring term.

Office hours

I will run an optional “office hours” drop-in session each week for feedback and consultation. You can come along if you want to talk to me about anything on the module, including if you’d like more feedback on your attempts at problem sheet questions. (For extremely short queries, you can approach me before or after lectures, but my response will often be: “Come to my office hours, and we can discuss it there!”)

Office hours will happen on **Thursdays from 1300 to 1400** – so directly after the Thursday lecture / problems class – in my office, which is **EC Stoner 9.10n** in “Maths Research Deck” area on the 9th floor of the EC Stoner building. (One way to the Maths Research Deck is via the doors directly opposite the main entrance to the School of Mathematics; you can also get there from Staircase 1 on the Level 10 “red route” through EC Stoner, next to the Maths Satellite.) If you cannot make this time, contact me for an alternative arrangement.

Exam

There will be one exam, which will make up 80% of your module mark.

The exam will be in the January 2026 exam period (11–22 January); the date and time will be announced in December. The exam will be in person and on campus.

The exam will last 2 hours and 30 minutes. The exam will consist of 4 questions, all compulsory. You will be allowed to use a permitted calculator in the exam.

Content of MATH5702

Necessary background

I will assume you have studied probability and statistics at the undergraduate level. Strong confidence and high proficiency in basic material is more important than very deep knowledge of more complicated topics. Experience suggests that confidence and proficiency at basic statistics is the most important factor in doing well at this module. (Last year, students with a strong background in statistics had a median score of a Distinction, while students with a weak background in statistics had a median score of a Fail.)

For Leeds undergraduates, MATH2701/2715 Statistical Methods is an official prerequisite, although confidence and proficiency in the more basic material of MATH1700/1710/1712 Probability and Statistics is probably more important.

Some knowledge I will assume:

- **Probability:** Basic rules of probability; random variables, both continuous and discrete; “famous” distributions (especially the normal, exponential, and continuous uniform distributions); expectation, variance, covariance, correlation; law of large numbers and central limit theorem.
- **Statistics:** Estimation of parameters; bias and error; sample mean and sample variance; confidence intervals.

This module will also include an material on Markov chains. I won’t assume any pre-existing knowledge of this, and I will introduce all new material we need, but students who have studied Markov chains before (for example in the Leeds module MATH2702 Stochastic Processes / MATH2750 Introduction to Markov Processes) may find a couple of lectures here are merely a reminder of things they already know.

A “Probability and statistics background: self-test” is available on Minerva. This test consists of 10 short numerical answer or multiple choice questions. I recommend taking this test to assess your background preparation for MATH5702 – but this is not compulsory, and it is not officially marked for credit.

I expect most students who are well prepared for MATH5702 should get full, or nearly full, marks on this test on the first attempt. If you score 6 or less, I recommend spending significant time revising your basic probability and statistics, and (if you are not on MSc Statistics) you may want to discuss with me whether you think MATH5702 is the right module for you.

The lectures will include examples using the **R** program language. The coursework and problem sheets will require use of R. The exam, while just a “pencil and paper” exam, will require understanding and writing short portions of R code. We will assume basic R capability – that you can enter R commands, store R objects using the `<-` assignment, and perform basic arithmetic with numbers and vectors. Other concepts will be introduced as necessary. If you

want to use R on your own device, I recommend downloading (if you have not already) the [R programming language](#) and the [program RStudio](#). (These lecture notes were written in R using RStudio.)

Syllabus

We plan to cover the following topics in the module:

- **Monte Carlo estimation:** definition and examples; bias and error; variance reduction techniques: control variates, antithetic variables, importance sampling. [9 lectures]
- **Random number generation:** pseudo-random number generation using linear congruential generators; inverse transform method; rejection sampling [7 lectures]
- **Markov chain Monte Carlo (MCMC):** [7 lectures]
 - Introduction to Markov chains in discrete and continuous space
 - Metropolis–Hastings algorithm: definition; examples; MCMC in practice; MCMC for Bayesian statistics
- **Resampling methods:** Empirical distribution; plug-in estimation; bootstrap statistics; bootstrap estimation [4 lectures]
- Frequently-asked questions [1 lecture]

Together with the 5 problems classes, this makes 33 lectures.

Book

The following book is strongly recommended for the module:

- J Voss, *An Introduction to Statistical Computing: A simulation-based approach*, Wiley Series in Computational Statistics, Wiley, 2014

The library has [electronic access to this book](#) (and two paper copies).

Dr Voss is a lecturer in the School of Mathematics and the University of Leeds, and has taught MATH5702 many times. *An Introduction to Statistical Computing* grew out of his lecture notes for this module, so the book is ideally suited for this module. My lectures will follow this book closely – specifically:

- Monte Carlo estimation: Sections 3.1–3.3
- Random number generation: Sections 1.1–1.4
- Markov chain Monte Carlo: Section 2.3 and Sections 4.1–4.3

- Resampling methods: Section 5.2

For a second look at material, for preparatory reading, for optional extended reading, or for extra exercises, this book comes with my highest recommendation!

Part I

Monte Carlo estimation

1 Introduction to Monte Carlo

A “Probability and statistics background: self-test” is available on Minerva. This test consists of 10 short numerical answer or multiple choice questions. I recommend taking this test to assess your background preparation for MATH5702 – but this is not compulsory, and it is not officially marked for credit.

Summary:

- Statistical computing is about solving statistical problems by combining human ingenuity with computing power.
- The Monte Carlo estimate of $\mathbb{E} \phi(X)$ is

$$\hat{\theta}_n^{\text{MC}} = \frac{1}{n} \sum_{i=1}^n \phi(X_i),$$

where $X_1, \dots, X_n$ are IID random samples from X .

- Monte Carlo estimation typically gets more accurate as the number of samples n gets bigger.

Read more: [Voss, *An Introduction to Statistical Computing*](#), Section 3.1 and Subsection 3.2.1.

2 Uses of Monte Carlo

Summary:

- The indicator $\mathbb{I}_A(x)$ function of a set A is 1 if $x \in A$ or 0 if $x \notin A$.
- We can estimate a probability $\mathbb{P}(X \in A)$ by using the Monte Carlo estimate for $\mathbb{E} \mathbb{I}_A(X)$.
- We can estimate an integral $\int h(x) \, dx$ by using a Monte Carlo estimate with $\phi(x) f(x) = h(x)$.

Read more: [Voss, *An Introduction to Statistical Computing*](#), Section 3.1 and Subsection 3.2.1.

3 Monte Carlo error I: theory

Summary:

- The Monte Carlo estimator is unbiased.
- The Monte Carlo estimator has mean-square error $\text{Var}(\phi(X))/n$, so the root-mean-square error scales like $1/\sqrt{n}$.
- The mean-square error can be estimated by S^2/n , where S^2 is the sample variance of the $\phi(X_i)$.

Read more: [Voss, *An Introduction to Statistical Computing*](#), Subsection 3.2.2.

4 Monte Carlo error II: practice

Summary:

- We can approximate confidence intervals for a Monte Carlo estimate by using a normal approximation.
- To get the root-mean-square error below ϵ we need $n = \text{Var}(\phi(X))/\epsilon^2$ samples.
- We can use a two-step process, where a small “pilot” Monte Carlo estimation allows us to work out how many samples we will need for the big “real” estimation.

Read more: [Voss, *An Introduction to Statistical Computing*](#), Subsections 3.2.2–3.2.4.

5 Control variate

Summary:

- Variance reduction techniques attempt to improve on Monte Carlo estimation making the variance smaller.
- If we know $\eta = \mathbb{E} \psi(X)$, then the control variate Monte Carlo estimate is

$$\hat{\theta}_n^{\text{CV}} = \frac{1}{n} \sum_{i=1}^n (\phi(X_i) - \psi(X_i)) + \eta.$$

- The mean-square error of the control variate Monte Carlo estimate is

$$\text{MSE}(\hat{\theta}_n^{\text{MC}}) = \frac{1}{n} \text{Var}(\phi(X) - \psi(X)).$$

Read more: [Voss, *An Introduction to Statistical Computing*](#), Subsection 3.3.3.

6 Antithetic variables I

Summary:

- Estimation is helped by combining individual estimates that are negatively correlated.
- For antithetic variables Monte Carlo estimation, we take pairs of non-independent variables (X, X') , to get the estimator

$$\hat{\theta}_n^{\text{AV}} = \frac{1}{n} \sum_{i=1}^{n/2} (\phi(X_i) + \phi(X'_i)).$$

On [Problem Sheet 1](#), you should now be able to answer all questions. You should work through this problem sheet in advance of the problems class on *Thursday 15 October* @ 11.

Read more: [Voss, *An Introduction to Statistical Computing*](#), Subsection 3.3.2.

Problem Sheet 1

This is Problem Sheet 1, which covers material from Lectures 1 to 6. You should work through all the questions on this problem sheet in advance of the problems class, which takes place in the lecture of **Thursday 15 October** @ 11.

This problem sheet is to help you practice material from the module and to help you check your learning. It is *not* for formal assessment and does not count towards your module mark.

However, if, optionally, you would like some brief informal feedback on **Questions 4, 6 and 8** (marked), I am happy to provide some. If you want some brief feedback, you should submit your work electronically through Gradescope via the module's Minerva page by **1400 on Tuesday 13 October**. (If you hand-write solutions on paper, the easiest way to scan-and-submit that work is using the Gradescope app on your phone.) I will return some brief comments on your those two questions by the problems class on Thursday 15 October. Because this informal feedback, and not part of the official assessment, I cannot accept late work for any reason – but I am always happy to discuss any of your work on any question in my office hours.

Many of these questions will require use of the [R programming language](#) (for example, by using the [program RStudio](#)).

Full solutions will be released on Friday 16 October.

A “Probability and statistics background: self-test” is available on Minerva. This test consists of 10 short numerical answer or multiple choice questions. I recommend taking this test to assess your background preparation for MATH5702 – but this is not compulsory, and it is not officially marked for credit.